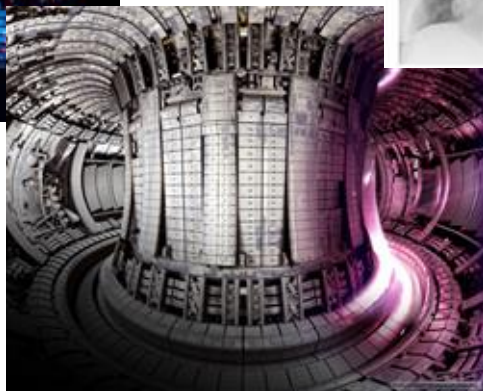
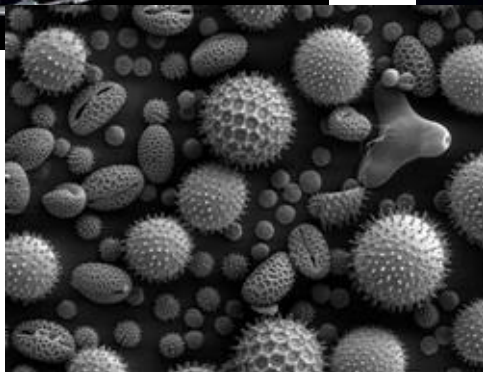
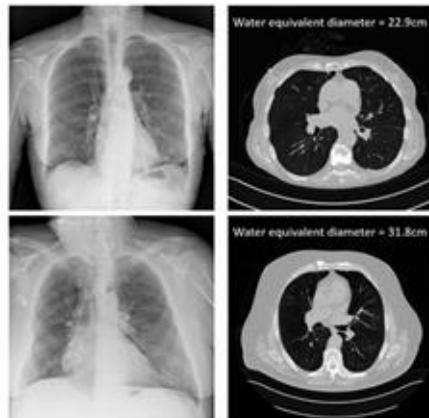
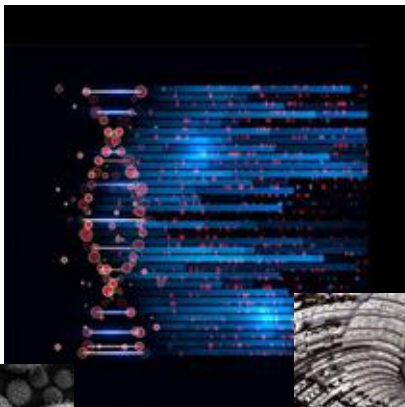
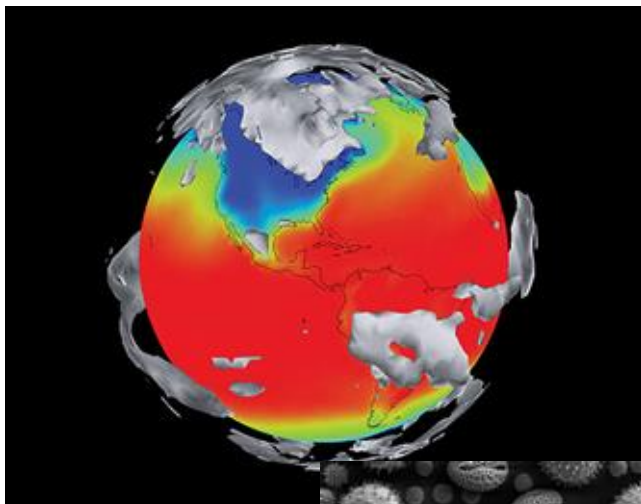


Intro to Big Array Data Analysis in Python

for geospatial science and microscopy

Multidimensional array data is everywhere



and more

What is Xarray?

Xarray is a cross-domain Python package that enables the analysis of large multidimensional datasets (mainly within the scientific domain). It does this in a few key ways:

- **introduces labels** in the form of **dimensions, coordinates, and attributes** on top of raw NumPy-like arrays, which allows for more intuitive, more concise, and less error-prone user experience
- **Integrates with lazy-array backends** (such as Dask) – allowing for the planning and execution of operations without exceeding RAM limitations
- Allows easy access to a range of n-dimensional data formats

Let's dive into Xarray!



The website for this workshop



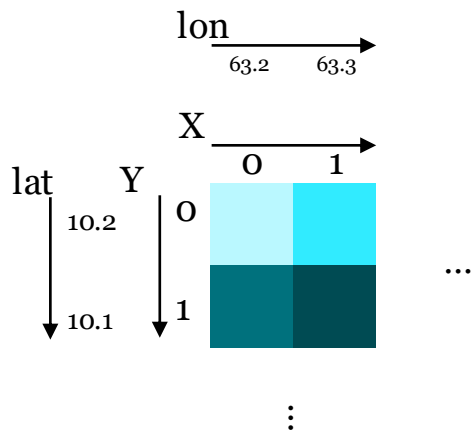
<https://uu-imau.github.io/xarray-UNECSS-2026/>

Tensors vs. tables

A fundamental distinction

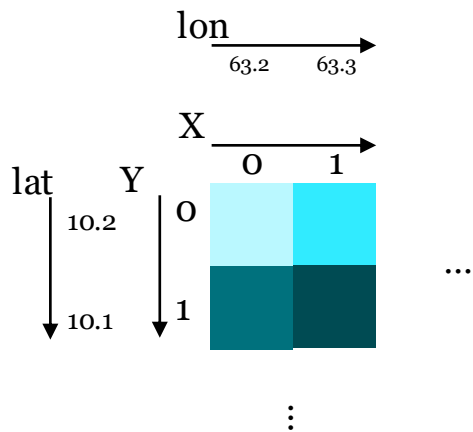
Tensors vs. tables

A fundamental distinction



Tensors vs. tables

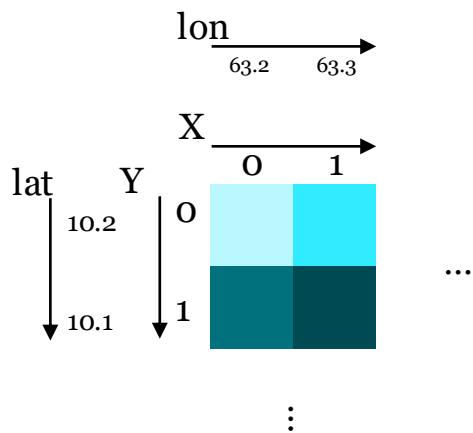
A fundamental distinction



X	Y	lon	lat	value

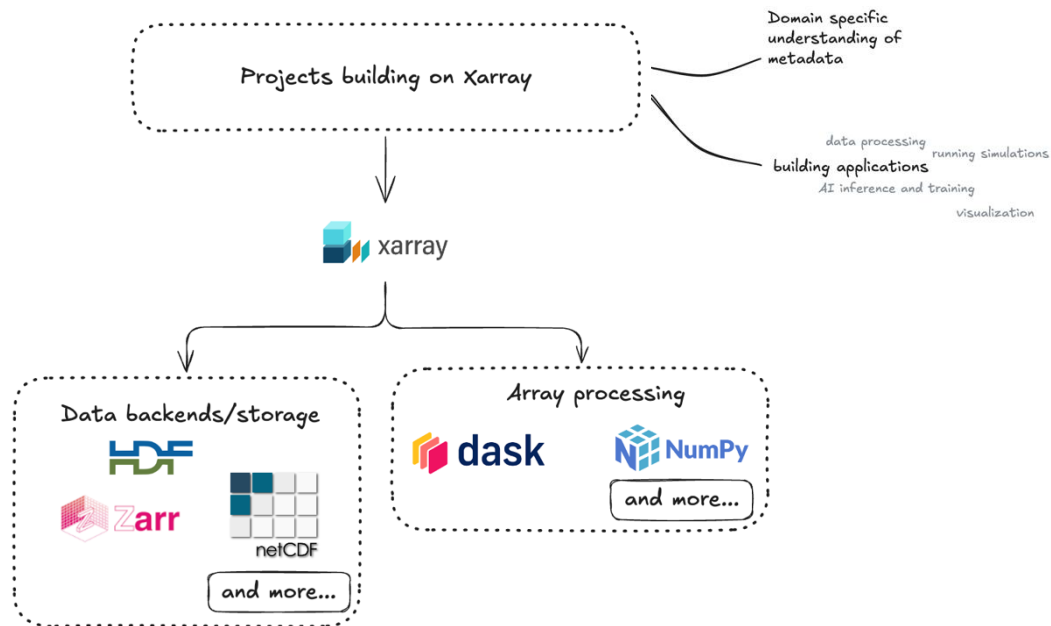
Tensors vs. tables

A fundamental distinction



X	Y	lon	lat	value
0	0	63.2	10.2	
1	0	63.3	10.2	
0	1	63.2	10.1	
1	1	63.3	10.1	

The ecosystem at a glance

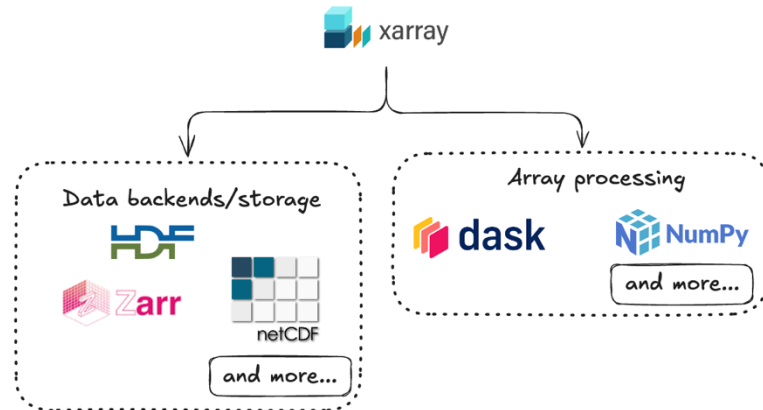


Where is it running?

Local machine

HPC machine

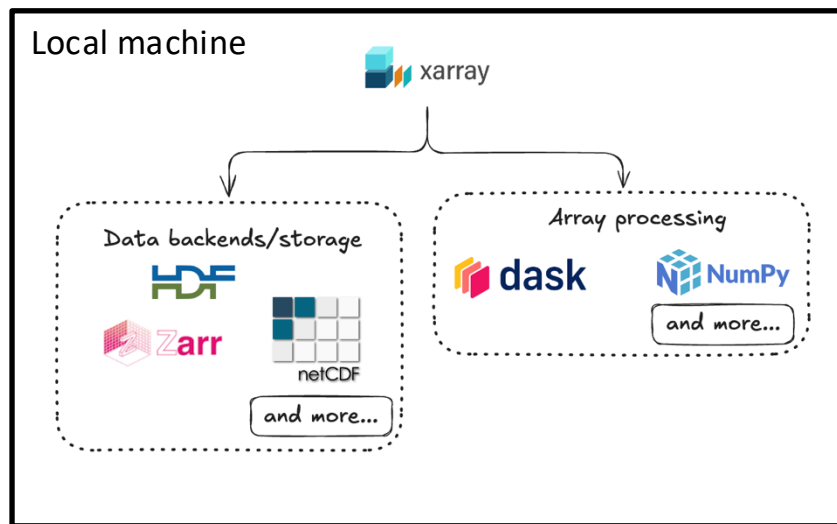
In the cloud



Where is it running? *All local*

HPC machine

In the cloud

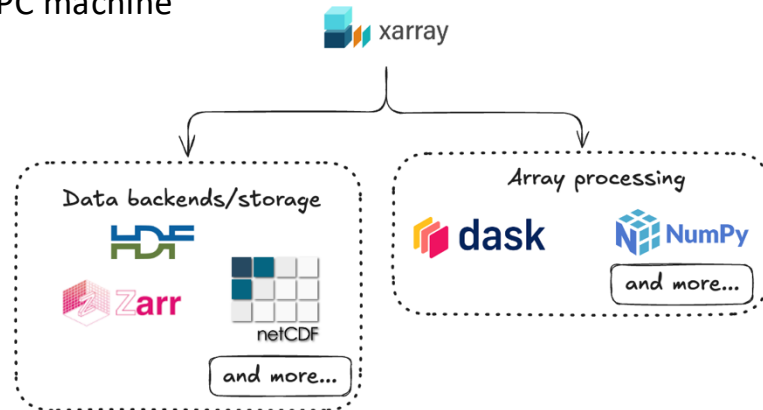


Where is it running? *All on HPC*

Local machine

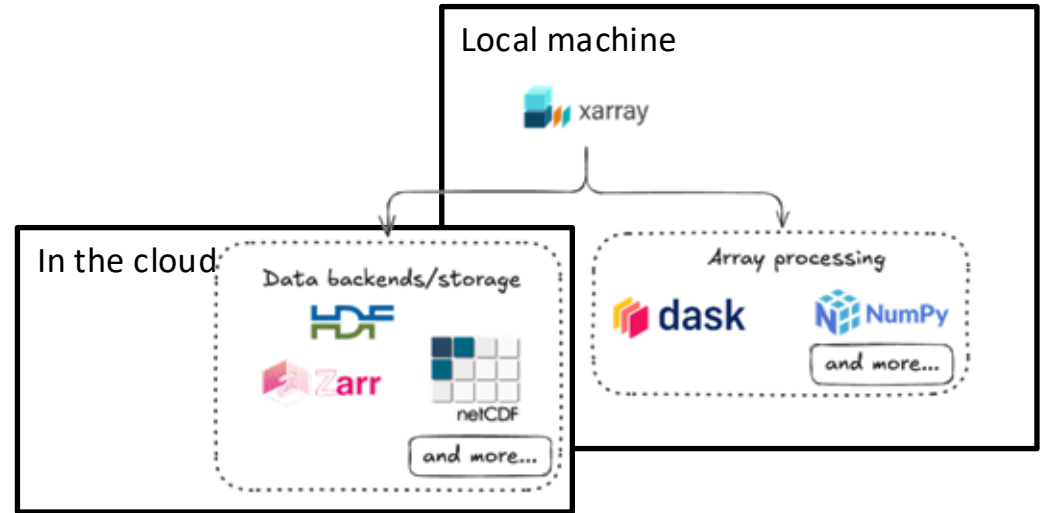
In the cloud

HPC machine



Where is it running? *Mix of local/HPC and cloud data*

HPC machine

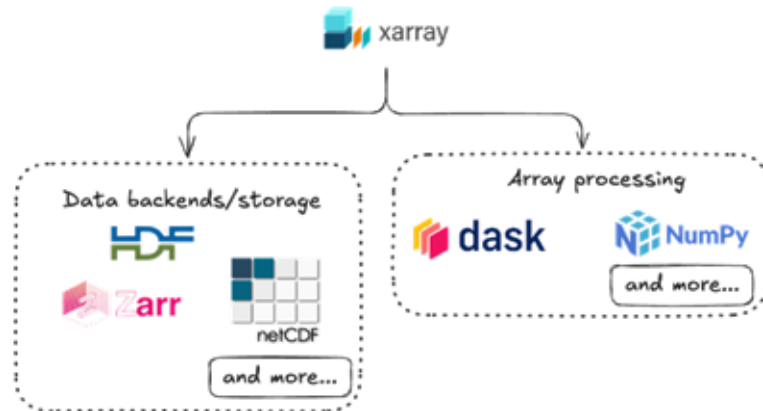


Where is it running? *All in the cloud*

Local machine

HPC machine

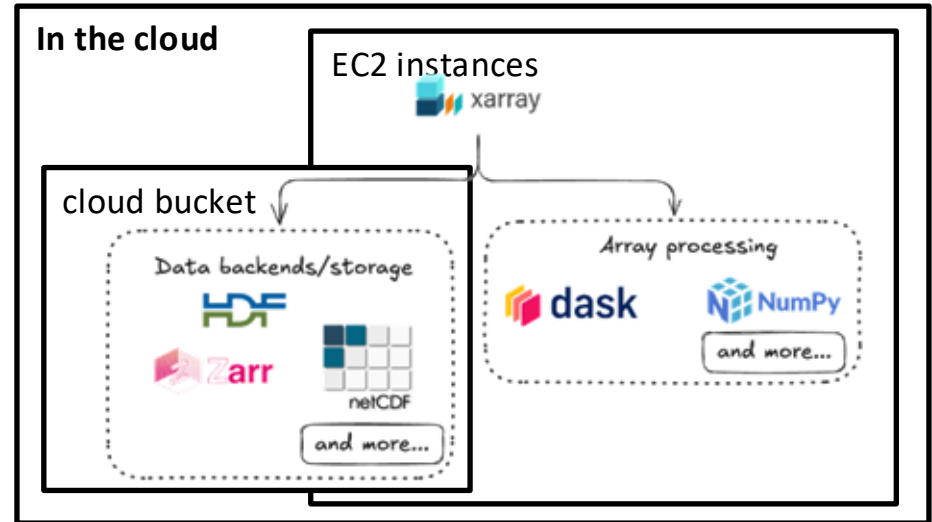
In the cloud



Where is it running? *All in the cloud - AWS*

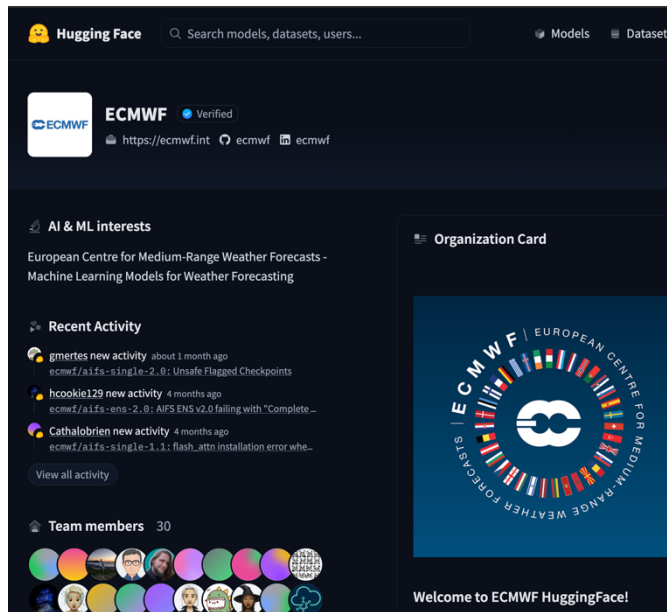
Local machine

HPC machine

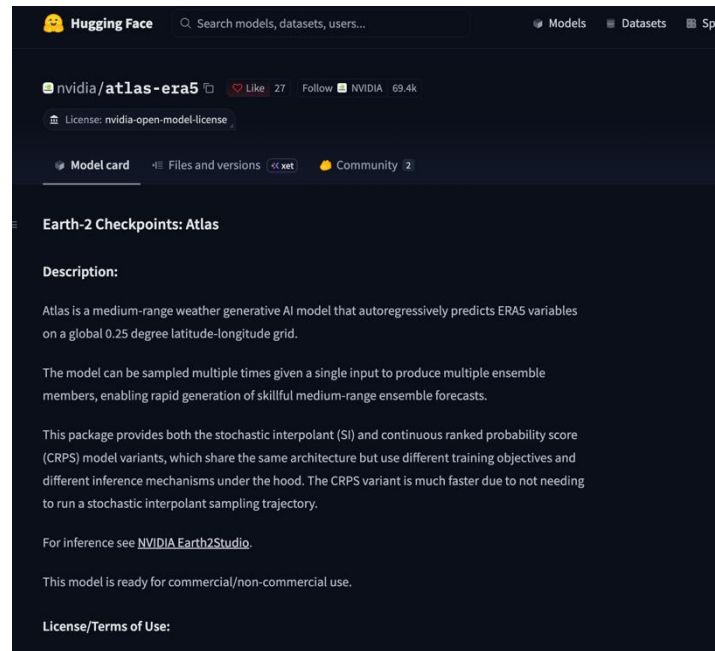


Through the lens of AI

AI models producing multidimensional data



<https://huggingface.co/ecmwf>



<https://huggingface.co/nvidia/atlas-era5>

Through the lens of AI

RESEARCH ARTICLE | ATMOSPHERIC SCIENCE



Physics-based models outperform AI weather forecasts of record-breaking extremes

ZHONGWEI ZHANG , ERICH FISCHER , JAKOB ZSCHEISCHLER , AND SEBASTIAN ENGELKE [Authors Info & Affiliations](#)

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26,567 6



Abstract

INTRODUCTION

RESULTS

DISCUSSION

MATERIALS AND METHODS

Acknowledgments

Supplementary Materials

REFERENCES

eLetters (0)

Abstract

Artificial intelligence (AI)-based models are revolutionizing weather forecasting and have surpassed leading numerical weather prediction systems on various benchmark tasks. However, their ability to extrapolate and reliably forecast unprecedented extreme events remains unclear. Here, we show that for record-breaking weather extremes, the physics-based numerical model High RESolution forecast (HRES) from the European Centre for Medium-Range Weather Forecasts still consistently outperforms state-of-the-art AI models GraphCast, GraphCast operational, Pangu-Weather, Pangu-Weather operational, and Fuxi. We demonstrate that forecast errors in AI models are consistently larger for record-breaking heat, cold, and wind than in HRES across nearly all lead times. We further find that the examined AI models tend to underestimate both the frequency and intensity of record-breaking events, and they underpredict hot records and overestimate cold records



<https://doi.org/10.1126/sciadv.aec1433>

Through the lens of AI

RESEARCH ARTICLE | ATMOSPHERIC SCIENCE

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26,567

Abstract

Artificial intelligence (AI)-based models are revolutionizing weather forecasting and have surpassed leading numerical weather prediction systems on various benchmark tasks. However, their ability to extrapolate and reliably forecast unprecedented extreme events remains unclear. Here, we show that for record-breaking weather extremes, the physics-based numerical model High RESolution forecast (HRES) from the European Centre for Medium-Range Weather Forecasts still consistently outperforms state-of-the-art AI models GraphCast, GraphCast operational, Pangu-Weather, Pangu-Weather operational, and Fuxi. We demonstrate that forecast errors in AI models are consistently larger for record-breaking heat, cold, and wind than in HRES across nearly all lead times. We further find that the examined AI models tend to underestimate both the frequency and intensity of record-breaking events, and they underpredict hot records and overestimate cold records

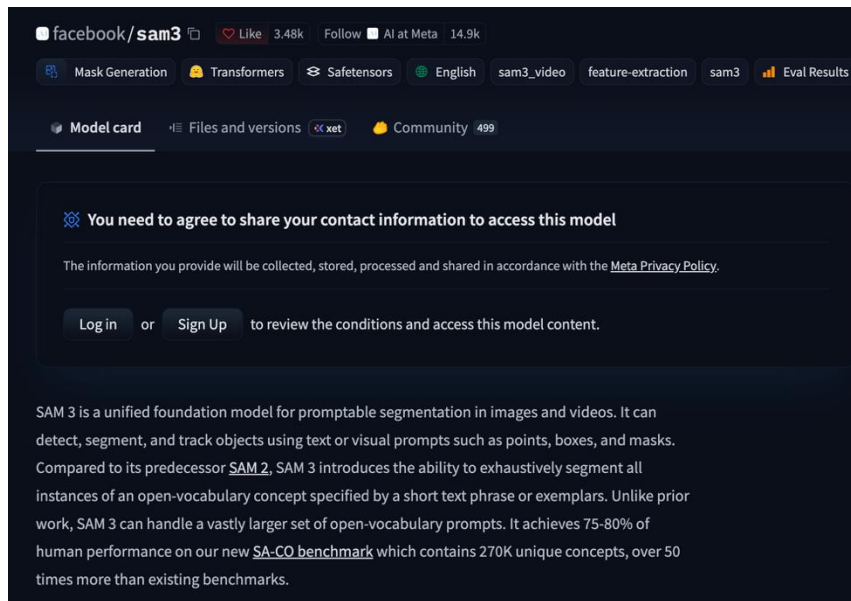
Abstract | INTRODUCTION | RESULTS | DISCUSSION | MATERIALS AND METHODS | Acknowledgments | Supplementary Materials | REFERENCES | eLetters (0)

- AI weather forecasting models are revolutionising weather forecasting.
- Running these models is cheap.

Through the lens of AI

AI models consuming multidimensional data

General image segmentation



The screenshot shows the Hugging Face model card for 'facebook/sam3'. At the top, it displays the model name, a 'Like' button with 3,48k likes, a 'Follow' button, and the creator 'AI at Meta' with 14.9k followers. Below this are tabs for 'Mask Generation', 'Transformers', 'Safetensors', 'English', 'sam3_video', 'feature-extraction', 'sam3', and 'Eval Results'. The 'Model card' tab is selected. A prominent message states: 'You need to agree to share your contact information to access this model'. Below this, it says: 'The information you provide will be collected, stored, processed and shared in accordance with the Meta Privacy Policy.' There are 'Log in' and 'Sign Up' buttons. The main text describes SAM 3 as a unified foundation model for promptable segmentation in images and videos, capable of detecting, segmenting, and tracking objects using text or visual prompts. It compares SAM 3 to its predecessor SAM 2, highlighting its ability to exhaustively segment all instances of an open-vocabulary concept and its performance on the SA-CO benchmark.

facebook/sam3 Like 3,48k Follow AI at Meta 14.9k

Mask Generation Transformers Safetensors English sam3_video feature-extraction sam3 Eval Results

Model card Files and versions xet Community 499

You need to agree to share your contact information to access this model

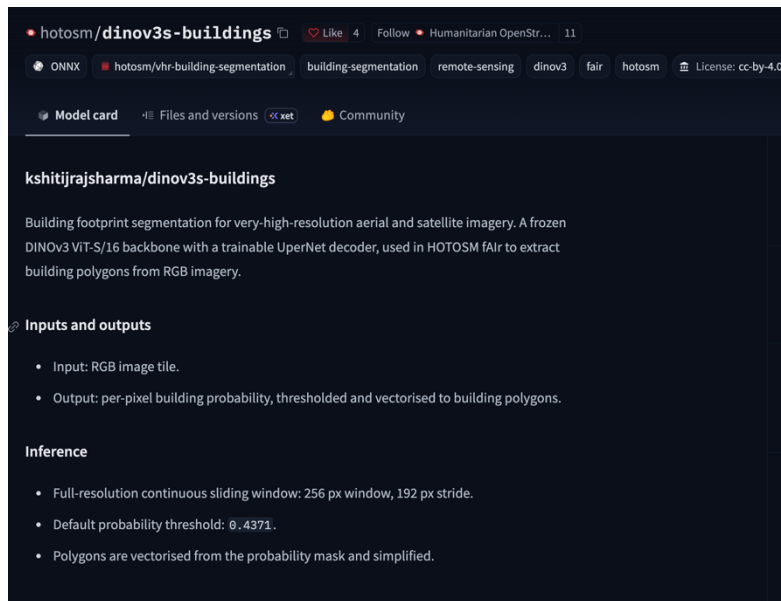
The information you provide will be collected, stored, processed and shared in accordance with the [Meta Privacy Policy](#).

Log in or Sign Up to review the conditions and access this model content.

SAM 3 is a unified foundation model for promptable segmentation in images and videos. It can detect, segment, and track objects using text or visual prompts such as points, boxes, and masks. Compared to its predecessor **SAM 2**, SAM 3 introduces the ability to exhaustively segment all instances of an open-vocabulary concept specified by a short text phrase or exemplars. Unlike prior work, SAM 3 can handle a vastly larger set of open-vocabulary prompts. It achieves 75-80% of human performance on our new **SA-CO benchmark** which contains 270K unique concepts, over 50 times more than existing benchmarks.

<https://huggingface.co/facebook/sam3>

Building footprint segmentation



The screenshot shows the Hugging Face model card for 'hotosm/dinov3s-buildings'. At the top, it displays the model name, a 'Like' button with 4 likes, a 'Follow' button, and the creator 'Humanitarian OpenStr...' with 11 followers. Below this are tabs for 'ONNX', 'hotosm/vhr-building-segmentation', 'building-segmentation', 'remote-sensing', 'dinov3', 'fair', 'hotosm', and 'License: cc-by-4.0'. The 'Model card' tab is selected. The main text describes the model as 'Building footprint segmentation for very-high-resolution aerial and satellite imagery. A frozen DINOv3 ViT-S/16 backbone with a trainable UperNet decoder, used in HOTOSM FAIR to extract building polygons from RGB imagery.' Below this is a section for 'Inputs and outputs' with two bullet points: 'Input: RGB image tile.' and 'Output: per-pixel building probability, thresholded and vectorised to building polygons.' There is also an 'Inference' section with three bullet points: 'Full-resolution continuous sliding window: 256 px window, 192 px stride.', 'Default probability threshold: 0.4371.', and 'Polygons are vectorised from the probability mask and simplified.'

hotosm/dinov3s-buildings Like 4 Follow Humanitarian OpenStr... 11

ONNX hotosm/vhr-building-segmentation building-segmentation remote-sensing dinov3 fair hotosm License: cc-by-4.0

Model card Files and versions xet Community

kshitijrajsharma/dinov3s-buildings

Building footprint segmentation for very-high-resolution aerial and satellite imagery. A frozen DINOv3 ViT-S/16 backbone with a trainable UperNet decoder, used in HOTOSM FAIR to extract building polygons from RGB imagery.

Inputs and outputs

- Input: RGB image tile.
- Output: per-pixel building probability, thresholded and vectorised to building polygons.

Inference

- Full-resolution continuous sliding window: 256 px window, 192 px stride.
- Default probability threshold: 0.4371.
- Polygons are vectorised from the probability mask and simplified.

<https://huggingface.co/hotosm/dinov3s-buildings>

Maturity of Xarray in various domains

- **Geosciences:** Very mature with lots of documentation, and good support for geoscience specific tools. The [Pythia Foundations course](#) from Project Pythia is an amazing resource for learning Python-based geoscientific workflows (it covers everything you need to know – and is my go-to recommendation for geoscientists). The Xarray documentation itself also have geoscience examples.
- **Microscopy:** Emerging support. [Napari](#) is rewriting their internals to use Xarray. Initiatives such as the [Open Microscopy Environment \(OME\)](#) are pushing the frontier on microscopy data in the cloud, and integration with Xarray. Currently, this intersection* suffers from poor documentation and tutorial resources – a gap that is being actively worked on (join the [Xarray Zulip](#) and chat with Ian Hunt-Isaak if you're interested here!).
- **Other domains:** The intersections* are not well defined or documented.

* i.e., the intersection between Xarray and using domain-specific file formats, and implementing domain specific workflows in Xarray